ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO TECHNICAL PROGRAMMING

PRACTICE EXAM 4

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Paper 2 style (design and problem solving, Arduino and C++). Section A: objective questions (30) · Section B: designing a smart-conveyor state machine with millis(), processing sensor data (averaging, map() and a temperature sensor), writing a Motor class and debugging a program (70).

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Arduino code must be written in the Arduino language (C/C++). For C++ programs, assume that #include <iostream>, #include <string> and using namespace std; are at the top.
6. Draw diagrams neatly. A non-programmable calculator may be used.
7. Write neatly and legibly. Indent your code so that it is easy to read.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The Arduino function that runs over and over again is …

A setup() B loop() C main() D start()

1.1.2 The statement that makes pin 13 an output is …

A pinMode(13, OUTPUT); B digitalRead(13);
C analogWrite(13); D Serial.begin(13);

1.1.3 The analogue-to-digital converter of an Arduino Uno has a resolution of …

A 8 bits (0–255). B 12 bits (0–4 095).
C 10 bits (0–1 023). D 16 bits (0–65 535).

1.1.4 The function analogWrite() produces …

A a true analogue voltage. B a PWM signal.
C a serial message. D an interrupt.

1.1.5 The statement used to send text from the Arduino to the computer is …

A Serial.println() B digitalWrite() C tone() D map()

1.1.6 A variable that stores a value from millis() should be of type …

A int B char C unsigned long D bool

1.1.7 The internal pull-up resistor of an input pin is switched on with …

A pinMode(pin, OUTPUT); B digitalWrite(pin, LOW);
C analogRead(pin); D pinMode(pin, INPUT_PULLUP);

1.1.8 An enum is used to …

A give names to a set of integer constants, such as states.
B store decimal numbers.
C read sensors.
D create loops.

1.1.9 A debugging tool that pauses the program at a chosen line is a …

A comment. B header. C library. D breakpoint.

1.1.10 An error that only shows while the program runs, such as dividing by zero, is a …

A syntax error. B runtime error. C logic error. D spelling error.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 The setup() function runs only once.

1.2.2 digitalRead() returns a value from 0 to 1 023.

1.2.3 A state machine keeps track of what the system is doing at this moment.

1.2.4 The value returned by millis() starts again from zero after about 49 days.

1.2.5 A constructor must have a return type.

1.2.6 Getters and setters give controlled access to private data.

1.2.7 The compiler always finds logic errors.

1.2.8 Averaging several sensor readings reduces the effect of noise.

1.2.9 An Arduino Uno pin can safely supply 1 A directly to a motor.

1.2.10 The map() function changes a value from one range to another.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	One of the modes a system can be in, in a state machine	A	library
1.3.2	Event that moves a system from one mode to another	B	noise
1.3.3	Small, random changes in a sensor reading	C	state
1.3.4	Makes an input read HIGH while its button is not pressed	D	getter
1.3.5	Ignoring the fast on-off bouncing of a switch	E	object
1.3.6	Member function that returns the value of private data	F	debouncing
1.3.7	Member function that changes private data after checking it	G	transition
1.3.8	An instance of a class	H	Serial Monitor
1.3.9	Window that shows the messages a sketch prints	I	setter
1.3.10	Collection of ready-made code that you can include	J	pull-up resistor

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: A SMART CONVEYOR STATE MACHINE

An Arduino controls a small conveyor in a robot work cell. Pressing the start button starts the conveyor. When the infrared sensor detects a box, the conveyor stops for 2 seconds so that a robot arm can pick up the box, and then it starts again. A green LED is on while the conveyor runs and a yellow LED is on while it waits at the station. The start button is on pin 2 (INPUT_PULLUP, LOW when pressed), the IR sensor on pin 3 (LOW when a box is detected), the conveyor relay on pin 8, the green LED on pin 9 and the yellow LED on pin 10.

2.1 List the inputs and the outputs of this system. (4)

2.2 Draw a state diagram for the conveyor with the states IDLE, RUNNING and AT_STATION. Label every transition with the event that causes it. (6)

2.3 Write the Arduino code for the conveyor. Use a variable for the state and millis() (not delay()) for the 2-second wait. Include the global variables, setup() and loop(). (10)

[20]

QUESTION 3: PROCESSING SENSOR DATA

3.1 Explain why a robot program often takes the average of several analogue readings instead of using a single reading. (2)

3.2 Write a function averageReading() that receives an analogue pin number, stores 10 readings of that pin in an array and returns their average as an integer. (6)

3.3 What value does map(512, 0, 1023, 0, 100) return? Explain your answer. (2)

3.4 An LM35 temperature sensor gives 10 mV for every °C and is connected to A0 of an Arduino Uno that uses a 5 V reference.

3.4.1 Write ONE statement that converts the reading in the variable reading into a temperature in °C, stored in a float called tempC. (3)

3.4.2 Calculate the temperature when reading has the value 61. (3)

[16]

QUESTION 4: A CLASS FOR A ROBOT MOTOR

4.1 Write a C++ class called Motor with: a private integer speed (0 to 255) and a private bool forward; a constructor that sets speed to 0 and forward to true; a setter setSpeed() that only accepts values from 0 to 255 (values outside the range must be limited to 0 or 255); a getter getSpeed(); a function reverse() that changes the direction; and a function display() that prints the speed and the direction. (12)

4.2 Write the code in main() that creates two Motor objects called leftMotor and rightMotor, sets both speeds to 200, reverses the right motor and displays both motors. (4)

4.3 Explain the difference between a class and an object, using the Motor class as an example. (2)

[18]

QUESTION 5: DEBUGGING

5.1 The program below should add the numbers 1 to 5 and display the total, but it contains FIVE errors. For EACH error, give the line number, state whether it is a syntax error or a logic error, and write the corrected code. (10)

```
 1  int main() {
 2    int total = 0
 3    for (int i = 1; i <= 5; i++); {
 4      total = total + i;
 5    }
 6    if (total = 15)
 7      cout << "Total is " << Total << endl;
 8    cout << "Done;
 9    return 0;
10  }
```

5.2 Explain the difference between a syntax error, a runtime error and a logic error. (3)

5.3 Name TWO techniques a programmer can use to find logic errors. (2)

5.4 Write down the output of the program once all the errors have been corrected. (1)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER